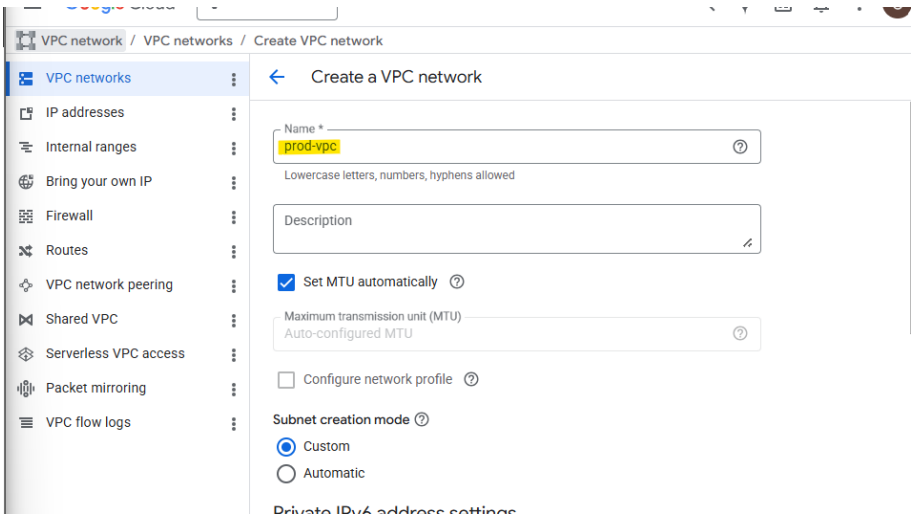
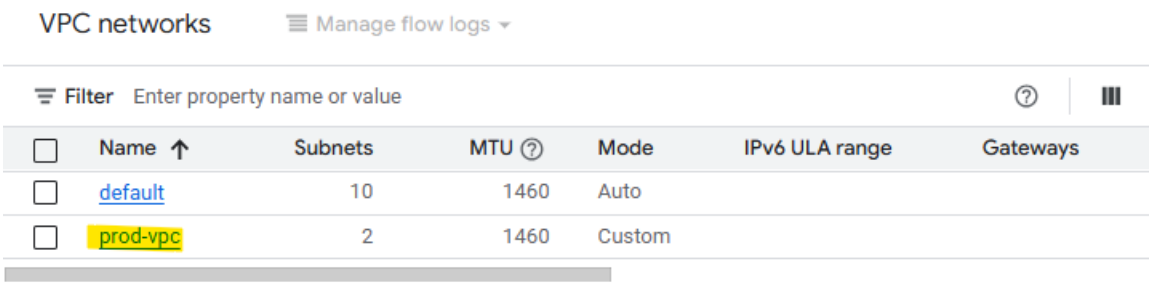
1. Project Setup

The project was pre-created in the Whizlabs sandbox.

2. Networking Layer (VPC Design)

Create a VPC Network

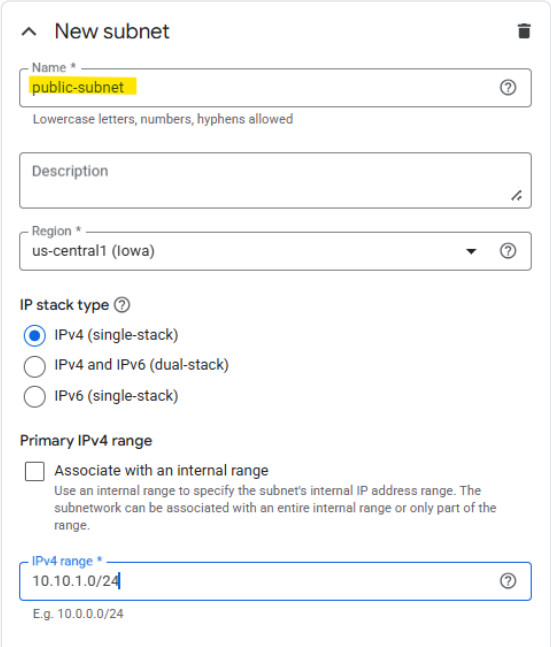




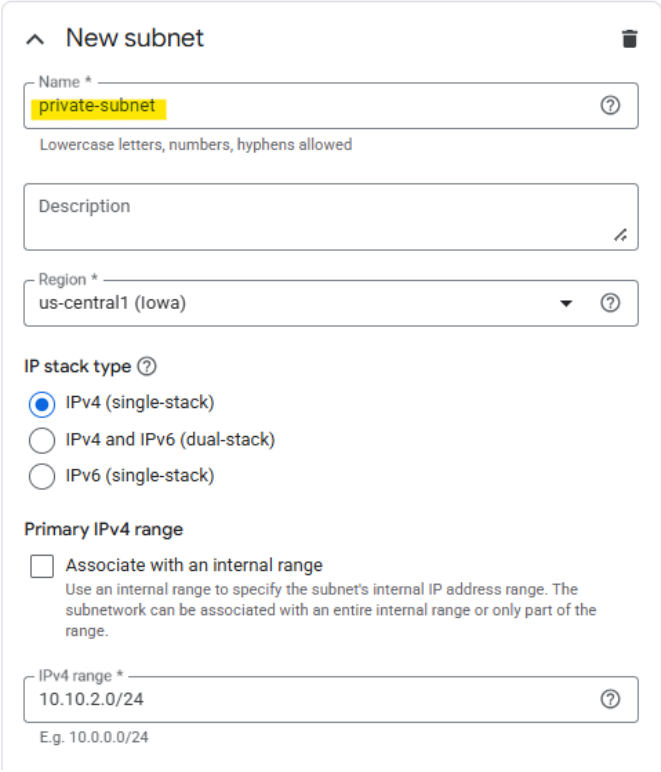
Name	Subnets	MTU	Mode	IPv6 ULA range	Gateways
default	10	1460	Auto		
prod-vpc	2	1460	Custom		

Create Subnets in different regions or zones:

Public Subnet



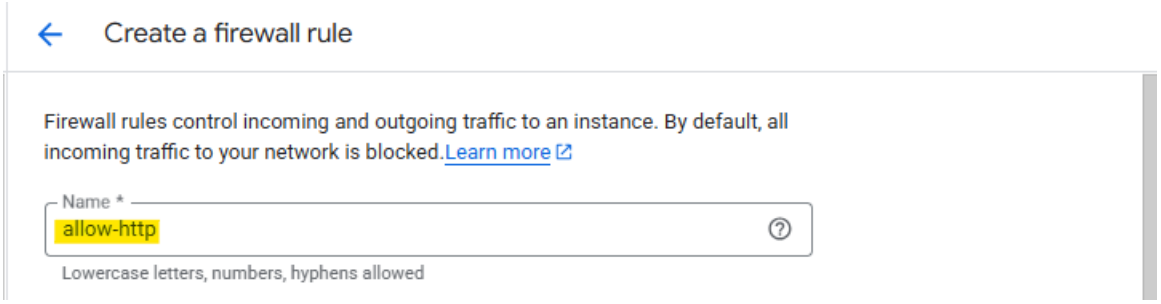
Private Subnet

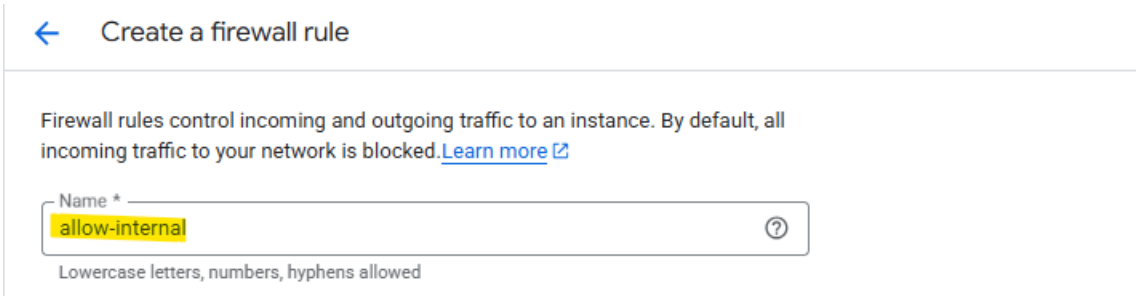


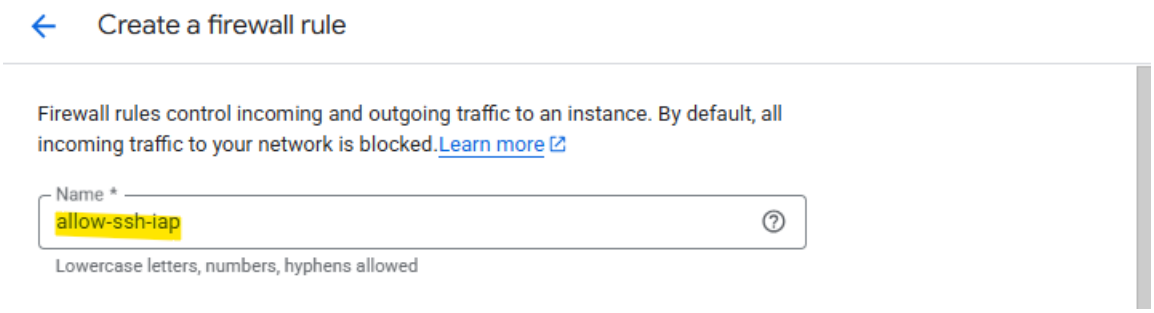
Routes

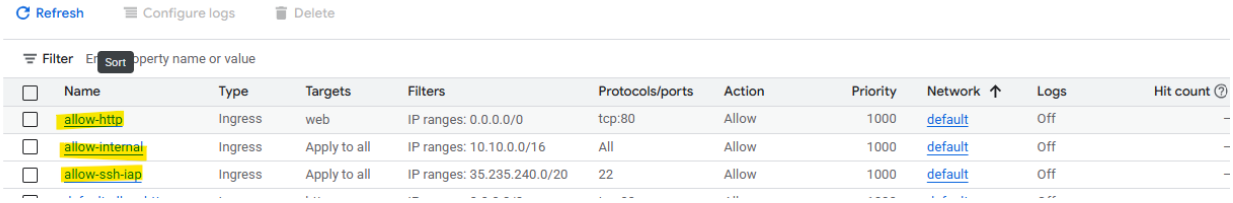
(Default routes are already created automatically)

Firewall Rules









Cloud NAT

Create Cloud NAT gateway

Cloud NAT lets your VMs and container pods create outbound connections to the Internet or to other virtual private cloud (VPC) networks.

Cloud NAT uses Cloud NAT gateway to manage those connections. Cloud NAT gateway is region and VPC network specific. If you have VM instances in multiple regions, you'll need to create a Cloud NAT gateway for each region. Learn more

Gateway name *
prod-nat

Lowercase letters, numbers, hyphens allowed

NAT type

- (●) Public
- ( ) Private

Select Cloud Router

Network *
prod-vpc

Region *
us-central1 (Iowa)

2 subnets.

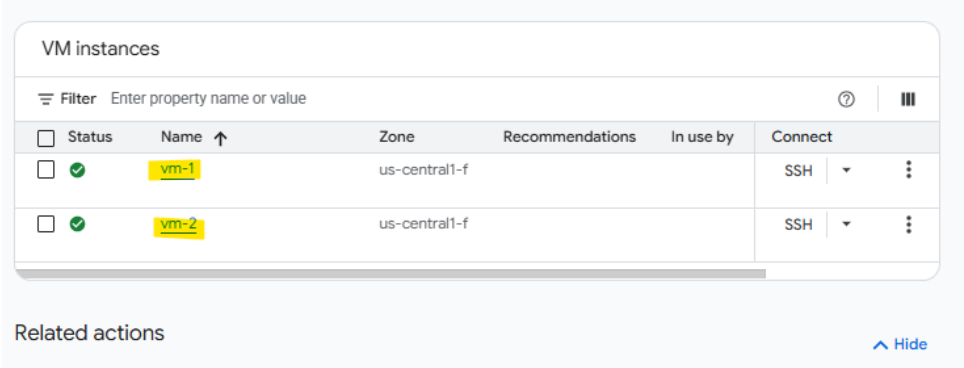
Cloud NAT | Create Cloud NAT gateway | Delete | Refresh | Learn

Filter Enter property name or value

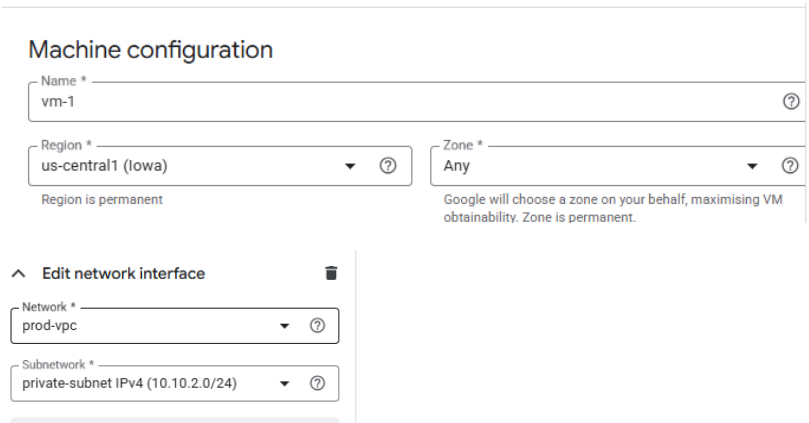
Gateway name	Network	Region	NAT type	Cloud Router
prod-nat	prod-vpc	us-central1	Public	prod-router

3. Compute Layer

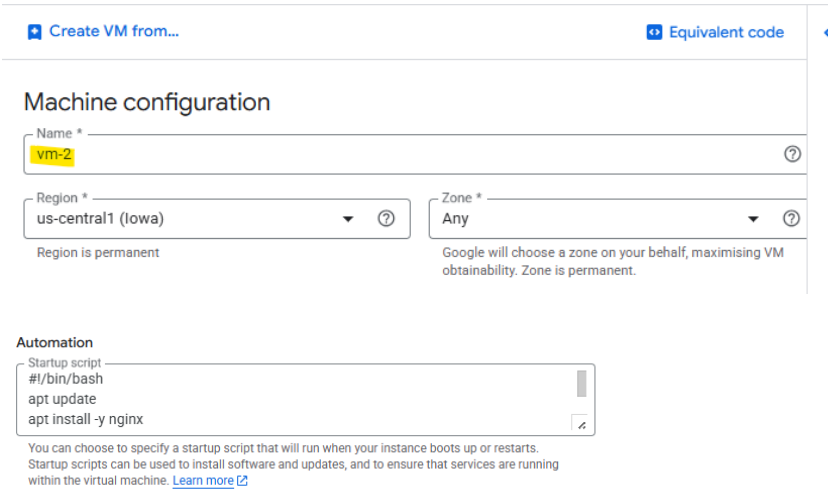
2 Virtual Machines (Compute Engine) in private subnet



VM1

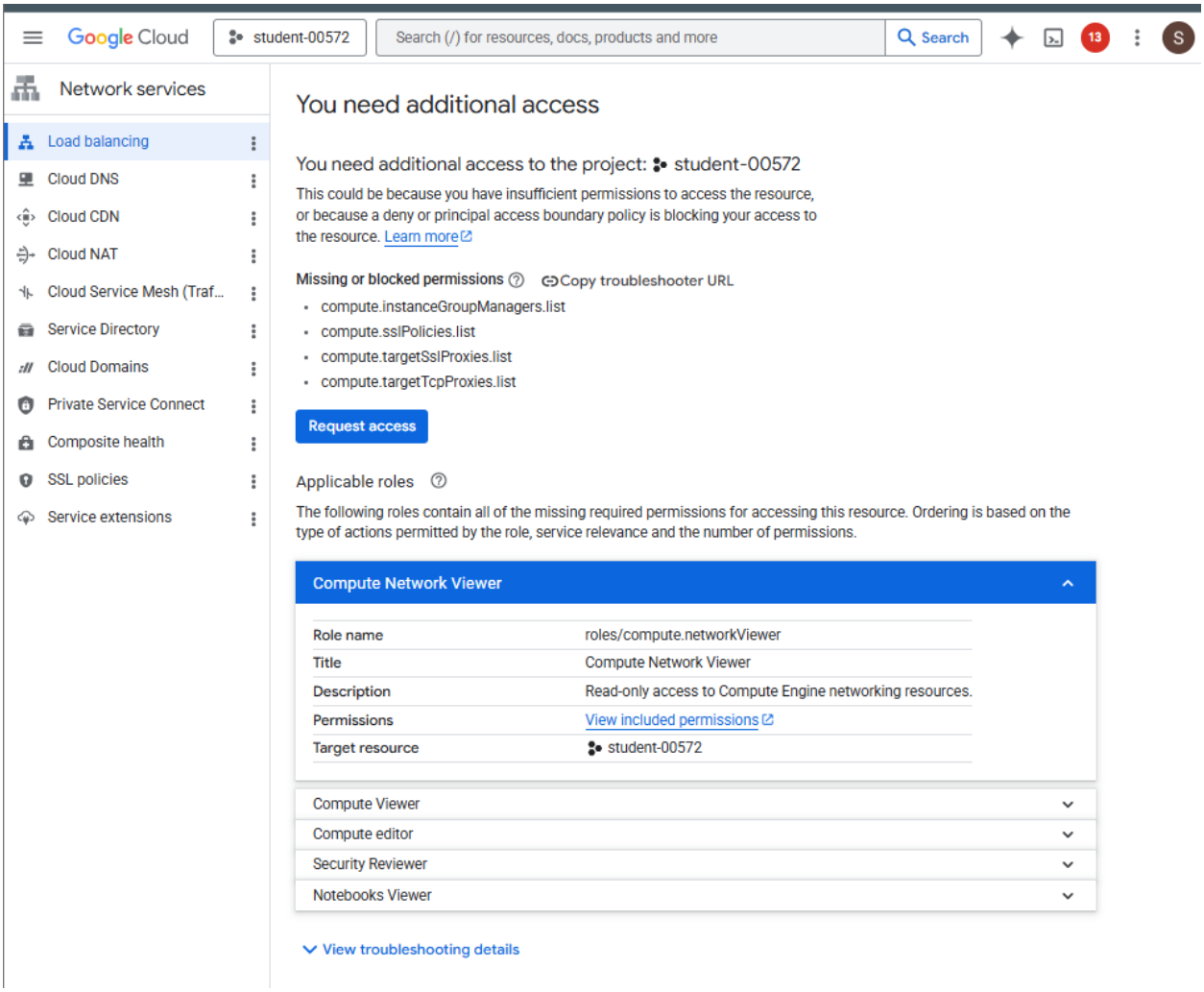


VM2



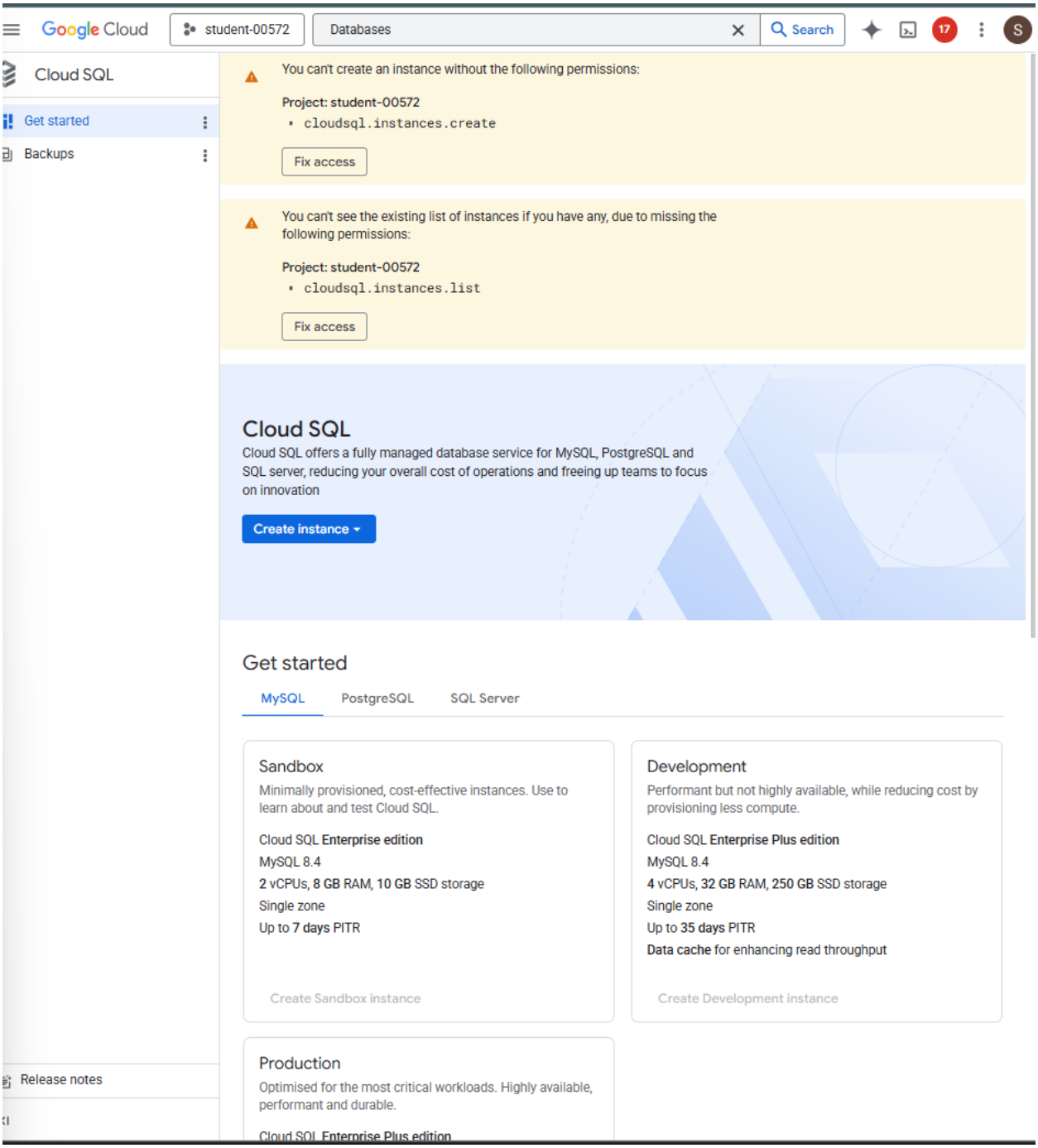
4. Load Balancing

I don’t have permission to create a Load Balancer in Whizlabs sandbox



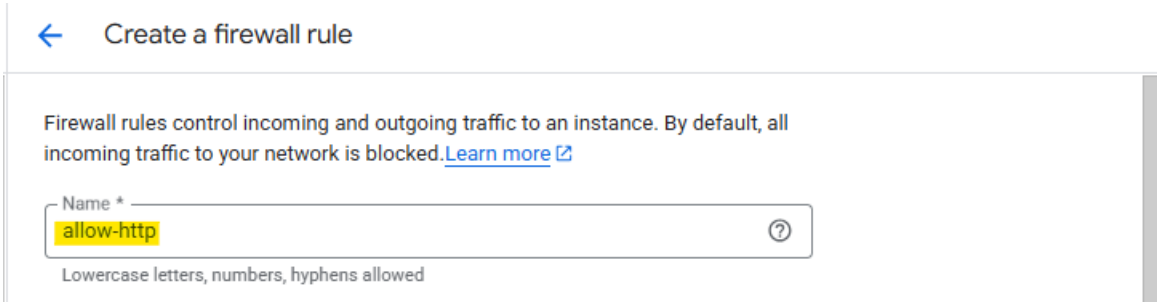
5. Data Layer

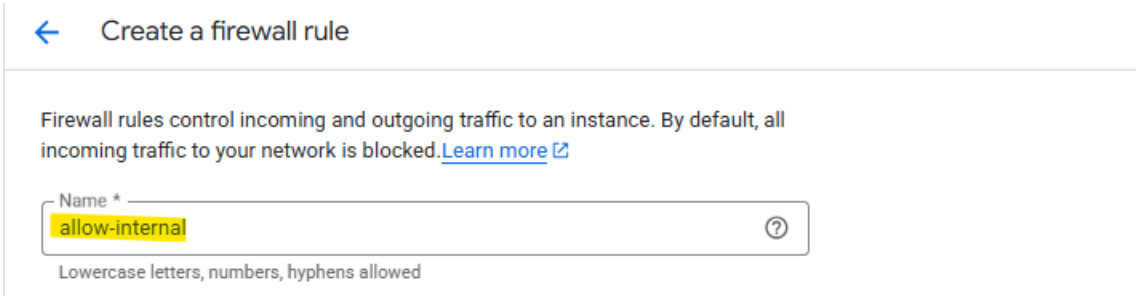
I don't have permission to create Cloud SQL in Whizlabs sandbox

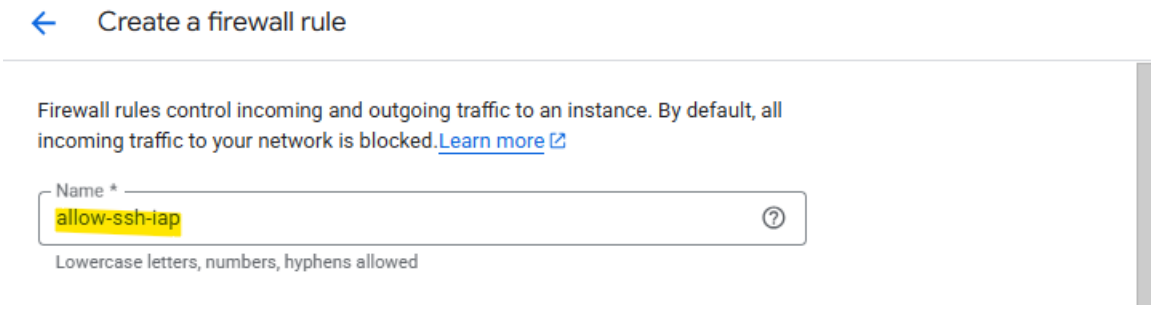


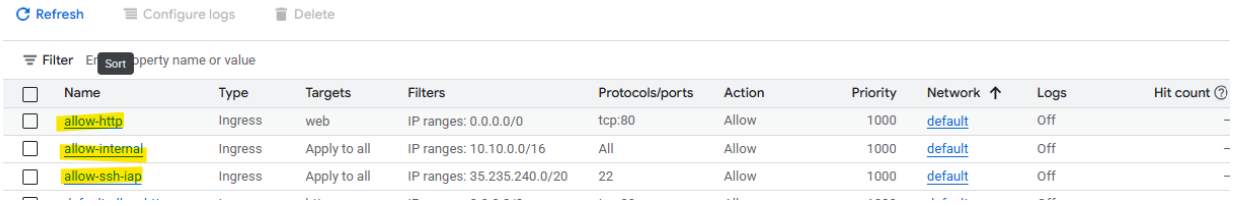
6. Security

Firewall Rules



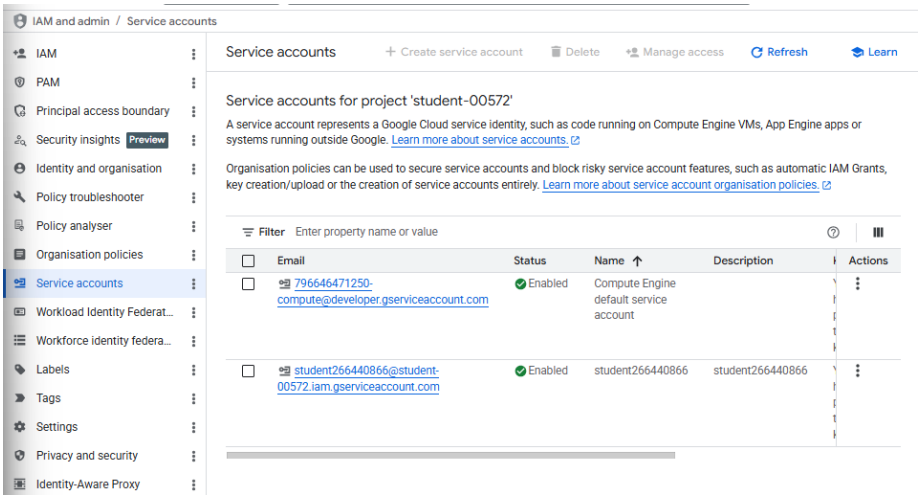






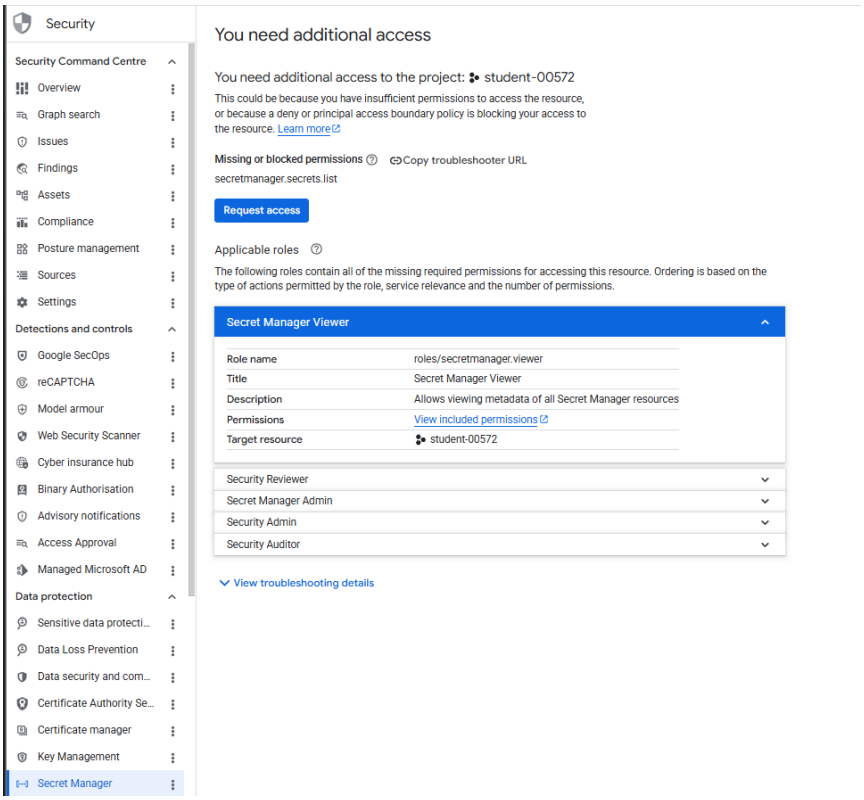
Service Account

I don’t have permission to create a new service account in Whizlabs sandbox



Secret Manager

I don’t have permission to create Secret Manager in Whizlabs sandbox



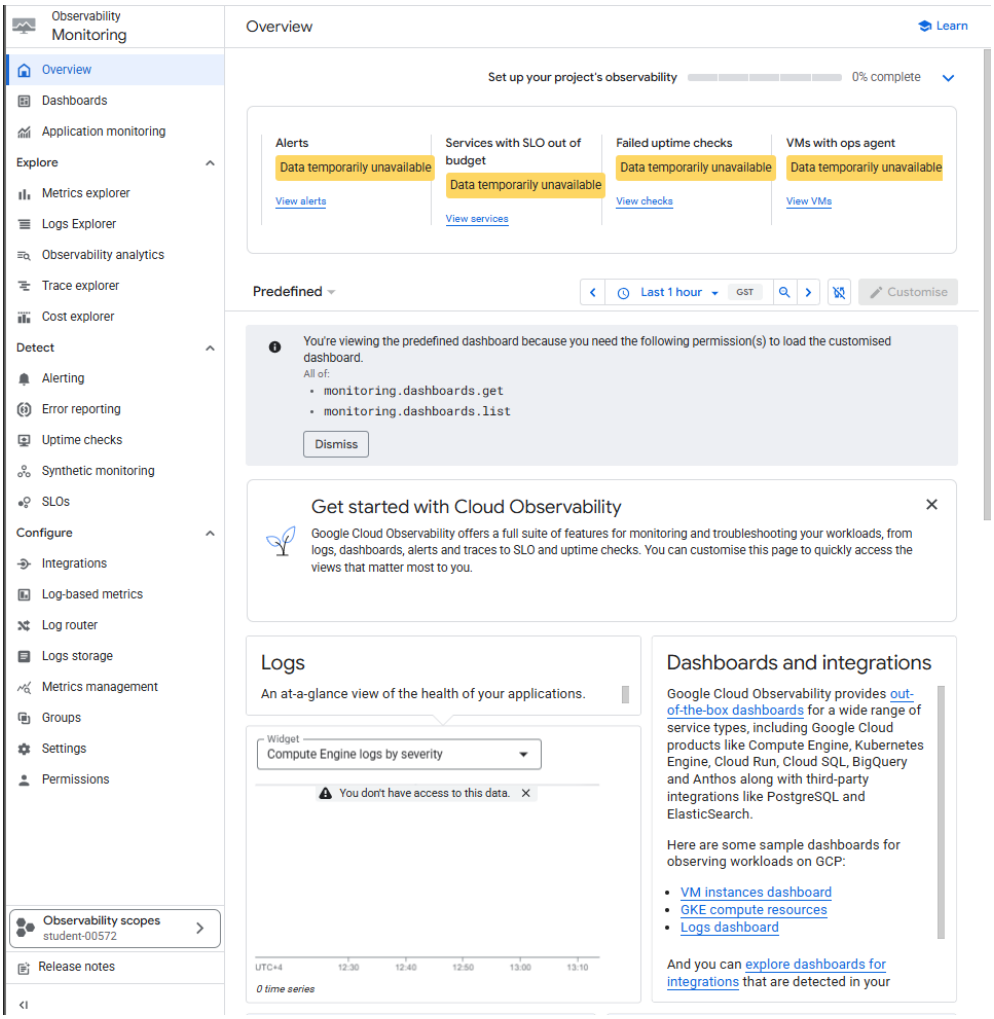
Least Privilege Principle

The least privilege principle was applied by assigning minimal permissions, restricting network access, and avoiding public exposure of resources.

7. Monitoring & Logging

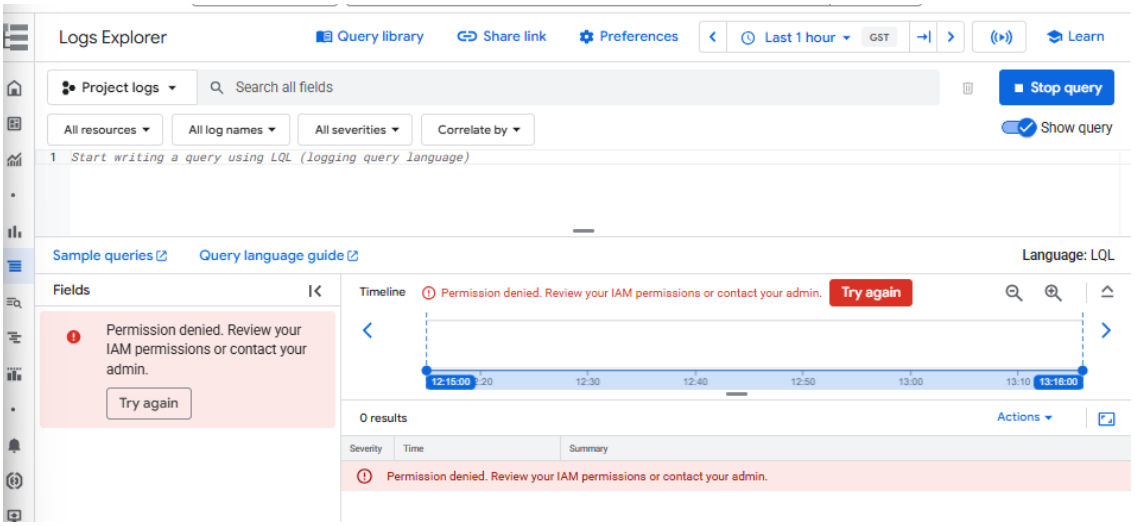
Cloud Monitoring

Cloud Monitoring was enabled; however, some metrics and dashboards were unavailable due to limited sandbox permissions.



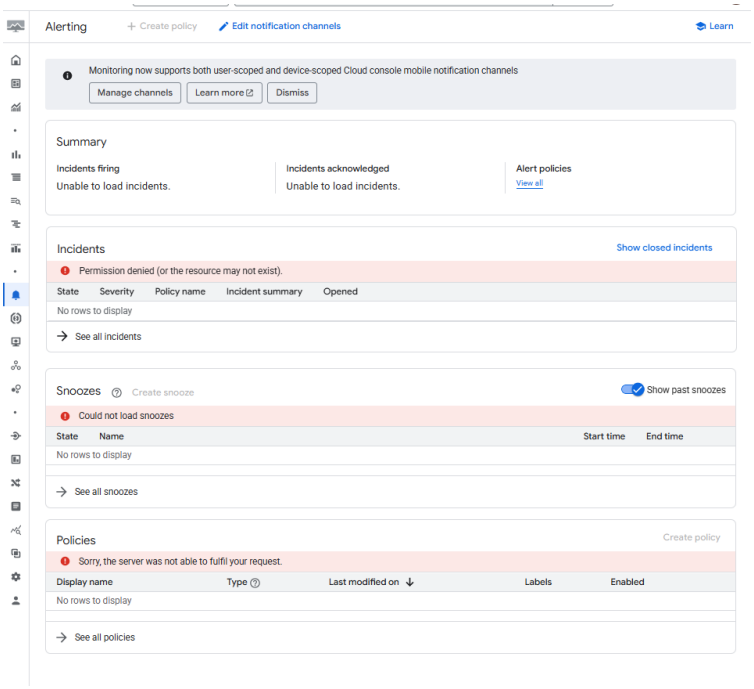
Cloud Logging

Cloud Logging was enabled; however, log data access was restricted due to limited IAM permissions in the sandbox environment.



Create Alert (CPU Usage)

access to alert policies and incidents was limited due to sandbox restrictions.



8. Automation (Mandatory)

```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
Your Cloud Platform project in this session is set to student-00572.
Use `gcloud config set project [PROJECT_ID]` to change to a different project.
student266440866@cloudshell:~ (student-00572)$ mkdir terraform-project
cd terraform-project
student266440866@cloudshell:~/terraform-project (student-00572)$ touch provider.tf main.tf
student266440866@cloudshell:~/terraform-project (student-00572)$ nano provider.tf
```

```
  GNU nano 7.2                                          provider.tf
provider "google" {
  project = "student-00572"
  region  = "us-central1"
}
```

```
  GNU nano 7.2                                            main.tf
resource "google_compute_network" "vpc" {
  name                    = "tf-vpc"
  auto_create_subnetworks = false
}

resource "google_compute_subnetwork" "subnet" {
  name          = "tf-subnet"
  ip_cidr_range = "10.10.2.0/24"
  region        = "us-central1"
  network       = google_compute_network.vpc.id
}

resource "google_compute_firewall" "allow_http" {
  name    = "tf-allow-http"
  network = google_compute_network.vpc.name

  allow {
    protocol = "tcp"
    ports    = ["80"]
  }

  source_ranges = ["0.0.0.0/0"]
}
```

```
student266440866@cloudshell:~/terraform-project (student-00572)$ terraform init

Initializing the backend...

Initializing provider plugins...
- Finding latest version of hashicorp/google...
- Installing hashicorp/google v7.28.0...
- Installed hashicorp/google v7.28.0 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
student266440866@cloudshell:~/terraform-project (student-00572)$
```

```
commands will detect it and remind you to do so if necessary.
student266440866@cloudshell:~/terraform-project (student-00572)$ terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with
the following symbols:
  + create

Terraform will perform the following actions:

  # google_compute_firewall.allow_http will be created
  + resource "google_compute_firewall" "allow_http" {
      + creation_timestamp = (known after apply)
      + destination_ranges = (known after apply)
      + direction          = (known after apply)
      + enable_logging     = (known after apply)
      + id                 = (known after apply)
      + name               = "tf-allow-http"
      + network            = "tf-vpc"
      + priority           = 1000
      + project            = "student-00572"
      + self_link          = (known after apply)
      + source_ranges      = [
          + "0.0.0.0/0",
        ]

      + allow {
          + ports    = [
              + "80",
```

```
Gemini CLI is available in Cloud Shell terminal. Type gemini to try it. Learn more        Don't show again   Dismiss

      + purpose                       = (known after apply)
      + region                        = "us-central1"
      + self_link                     = (known after apply)
      + stack_type                    = (known after apply)
      + state                         = (known after apply)
      + subnetwork_id                 = (known after apply)
    }

Plan: 3 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

google_compute_network.vpc: Creating...
google_compute_network.vpc: Still creating... [10s elapsed]
google_compute_network.vpc: Still creating... [20s elapsed]
google_compute_network.vpc: Still creating... [30s elapsed]
google_compute_network.vpc: Creation complete after 32s [id=projects/student-00572/global/networks/tf-vpc]
google_compute_firewall.allow_http: Creating...
google_compute_subnetwork.subnet: Creating...
google_compute_firewall.allow_http: Still creating... [10s elapsed]
google_compute_subnetwork.subnet: Still creating... [10s elapsed]
google_compute_firewall.allow_http: Creation complete after 11s [id=projects/student-00572/global/firewalls/tf-allow-
http]
google_compute_subnetwork.subnet: Still creating... [20s elapsed]
google_compute_subnetwork.subnet: Creation complete after 23s [id=projects/student-00572/regions/us-central1/subnetwo
rks/tf-subnet]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.
student266440866@cloudshell:~/terraform-project (student-00572)$
```

VPC networks Create VPC network Refresh Learn

Networks in current project Subnets in current project

SMTP port 25 allowed in this project. Learn more

VPC networks Manage flow logs

Filter Enter property name or value

Name	Subnets	MTU	Mode	IPv6 ULA range	Gateways	Flow log configs
default	10	1460	Auto			
prod-vpc	2	1460	Custom			
tf-vpc	1	1460	Custom			